Solving Single- and Multi-Lens Problems

Suppose you have an object at a specified position to the left of an optical device. Finding the properties of the resulting image is straightforward, though you do have to approach the problem systematically in order not to go wrong. It's as simple as that.

Single-Lens Problems

If the optical device is a single lens of specified focal length, then given the object position, this is a trivial problem to solve by using the Thin Lens Equation. Simply reorganize the Thin Lens Equation to compute *q* as a function of *p* and *f*, and then plug in values for *p* and *f*.

- The value of *q* is the image location, where a positive value means the image is to the right of the lens (and negative to the left). Light can only be focused by the lens on the right, so the image is real only if $q > 0$ (and virtual if $q < 0$).
- The magnification of the image relative to the object is $M = -q/p$. A positive value means the image is upright, while a negative value means it is inverted (upside down). The textbook characterizes images as reduced or enlarged depending upon whether the magnitude of *M* is less than or greater than 1.

Two-Lens Problems

The way to solve all two-lens problems is to break them down into sequential steps, following the progress of light through each lens, one at a time. As long as you do this, all will be well. To avoid confusion, take care to use different symbols to refer to the object and image distances from each lens. That is, p_1, q_1, f_1 for the first lens and p_2, q_2, f_2 for the second.

First, light has to travel through the left lens, so apply the above process to find q_1. The image formed by the first lens then serves as the object for the second lens, so you just apply the above process to the second lens given its object's location p_2.

- For the first lens, the single-lens solution tells you the location of the intermediate image,q_1.
- The intermediate image serves as the object for the second lens. Here is the important point: If the two lenses are separated by a distance *d*, then $p_2 = d - q_1$. Draw a picture of two lenses a distance *d* apart to convince yourself that this makes sense in all three cases:
    - $q_1 < 0 \; so \; p_2 > d$, and we have a real object for the second lens, or
    - $0 < q_1 < d \; so \; 0 < p_2 < d$, and we have a real object for the second lens, or
    - $q_1 > d \; so \; p_2 < 0$ and the intermediate image is to the right of the second lens. In this last case, we call this a *virtual object* for the second lens because light rays are going *toward* that location, not coming from it. In between the lenses, the light rays point in the direction as if it were there, even though the second lens will bend those rays before reaching that location.

Given the position of the intermediate image/object, solve the second lens problem exactly as you did the first. Easy and straightforward, as long as you approach it systematically.